USER MANUAL

ICS / OT / IoT Threat Modeler

A guided, model-agnostic threat-modeling tool for industrial & embedded systems

Single-file edition · Version 3.0

MITRE EMB3D™ · MITRE ATT&CK® for ICS

ISA/IEC 62443 (3-3 · 4-2 · 4-1 · 2-1)

ISA-TR84.00.09 · Attify IoT attack surface

Choose your AI model → name a component → upload artifacts → Analyze.
The AI produces the full report, mapping tables, and diagrams as one PDF.

Contents

1. **Introduction**
   - What the tool does
   - Who it is for
   - Frameworks covered
2. **How it works & data privacy**
3. **System requirements (nothing to install)**
4. **Installation & first run**
5. **Getting an API key (or running key-free with a local model)**
6. **Step 1 — Choose your AI model & key**
7. **Steps 2–4 — Describe, Artifacts, Analyze**
   - Step 2 — Describe
   - Step 3 — Artifacts
   - Step 4 — Analyze
8. **What the AI does automatically**
9. **Understanding the results**
10. **The diagrams (threat model, ICS kill chain, attack mind map)**
11. **The PDF report**
12. **Refining the analysis (updates report & PDF)**
13. **Saved reports, new models & comparing components**
14. **Command-line & environment reference**
15. **Troubleshooting**
16. **Frameworks, attribution & disclaimer**
17. **Frequently asked questions**
18. **Quick-reference card**

1. Introduction

1.1 What the tool does

The **ICS/OT/IoT Threat Modeler** builds rigorous, standards-aligned security threat models for industrial control systems, operational technology, and IoT devices — from a single PLC, RTU, or smart meter to a full control system including its sub-components (SoC, FPGA, radios, memories).

The workflow is deliberately simple: you **name the component or system**, optionally **upload supporting artifacts** (datasheets, manuals, P&IDs, design specs), and click **Analyze**. The AI then does the entire analysis on the back end — against every embedded framework and your uploaded material — and produces a narrative summary, mapping tables, and three diagrams, all collected into a single downloadable PDF report.

The whole application is **one Python file**. There is nothing to install: no packages, no build step, no cloud account. It runs on the standard library alone, so it also works on isolated or air-gapped engineering networks.

1.2 Who it is for

- **Systems security engineers** creating device/product threat models for design reviews and testing.
- **Product security & development teams** mapping designs to IEC 62443-4-1/4-2 for procurement and certification.
- **Asset owners & OT engineers** assessing deployed equipment against 62443-3-3/2-1.
- **Functional-safety practitioners** examining how cyber attacks could defeat safety functions (ISA-TR84.00.09).
- **Researchers & penetration testers** planning device teardown and attack-surface analysis (Attify methodology).

You do **not** need to be an expert in any of these frameworks to use the tool — selecting the relevant properties, threats, techniques, and requirements is the AI's job, not yours.

1.3 Frameworks covered

The complete reference data for the following frameworks is embedded in the tool (560 records) and is fed to the AI on every run so the analysis uses the frameworks' exact identifiers and titles.

Framework	What is embedded
MITRE EMB3D™	Device property types (PID), the full threat catalogue (TID), and the full mitigation catalogue (MID).
MITRE ATT&CK® for ICS	All 12 tactics, the full technique set (T0xxx), and all mitigations (M0xxx).
ISA/IEC 62443	3-3 system requirements (SR), 4-2 component requirements (CR + EDR/HDR/NDR/SAR), 4-1 lifecycle practices, and 2-1 asset-owner program elements.

ISA-TR84.00.09	Safety-lifecycle / cybersecurity concepts linking cyber attack paths to Safety Instrumented Systems (SIS/SIF), plus a consequence-severity scale.
Attify IoT	Reverse-engineering / attack-surface mindmap across hardware teardown, firmware, radio, network, and application testing.

> **Hybrid grounding.** The AI maps against this embedded core using exact IDs. Where a relevant item is not in the core, it may add well-known correct entries and marks them with a trailing asterisk (e.g. `TID-###*`) so you can tell model-supplied items from the reference set.

2. How it works & data privacy

The tool runs a small web server on your own machine (or a VM you control). Your browser talks to that local server; the server calls whichever AI provider you chose to perform the analysis. Nothing is sent to any third party except the AI provider you explicitly select.

The flow of one analysis

1. You enter a target, optional scope, and optional artifacts in the browser.
2. The local server extracts text from your artifacts, prepares any diagram images, and assembles a prompt that also contains the full embedded framework reference data.
3. The server sends that prompt (and any images) to your chosen AI provider using your API key.
4. The AI returns a structured result. The server renders it into the on-screen report, the three diagrams, and a downloadable PDF — and stores it locally so you can reopen it later.

> **Where your data goes.** Your target descriptions, artifacts, and generated reports stay on your machine (saved reports live in a local database file). The report view, tables, and all three diagrams are rendered entirely by the tool — no external web content is required to view a result. The only outbound traffic is the request to the AI provider you pick. API keys are stored only in your browser and/or read from environment variables you set; they are never written to the saved-reports database.

> **Handle sensitive material appropriately.** When you Analyze, your prompt — including any artifact text and images you attach — is sent to the AI provider you selected. Follow your organization's data-handling policy before submitting proprietary datasheets or design specifications to a third-party model, or use a local model (Section 5.3) to keep everything on-premises.

3. System requirements (nothing to install)

Operating system	Windows, macOS, or Linux (anything that runs Python 3).
Python	Version 3.9 or newer. Check with `python3 --version`.
Dependencies	**None.** The tool uses only the Python standard library — including reading PDF/Word artifacts and generating the PDF report. There is nothing to `pip install`.
Browser	Any modern browser (Chrome, Edge, Firefox, Safari).
Network	Outbound access to your chosen AI provider. A local model (Ollama, LM Studio) can run fully offline.
AI access	An API key for one provider, *or* a local/OpenAI-compatible endpoint. See Section 5.

> **Air-gap friendly.** Because there are no third-party dependencies and the report/diagrams render locally, the tool is well suited to isolated OT/engineering networks. Pair it with a local model to keep every part of the workflow on-premises.

4. Installation & first run

There is nothing to install beyond Python itself. The entire tool is the single file `ics_threat_modeler.py`.

Step 1 — Confirm Python

```
python3 --version       # should print 3.9 or newer
```

Step 2 — Start the tool

Open a terminal in the folder containing the file and run:

```
python3 ics_threat_modeler.py --open
```

The tool starts a local server and prints a banner similar to:

```
================================================================
  ICS/OT/IoT Threat Modeler  v3.0.0
  Embedded framework records: 560
  Serving at:  http://127.0.0.1:8000
  Saved reports DB:  /home/you/.ics_threat_modeler.db
  Press Ctrl+C to stop.
================================================================
```

With `--open`, your browser opens automatically. Otherwise browse to **http://127.0.0.1:8000**.

> **Windows note.** If `python3` is not recognized, use `python ics_threat_modeler.py --open`.

Step 3 — Stop the tool

Press **Ctrl+C** in the terminal. Your saved reports remain in the local database for next time.

Running on a VM or shared host

```
python3 ics_threat_modeler.py --host 0.0.0.0 --port 8000
```

> **Exposing the server.** Binding to `0.0.0.0` makes the UI reachable by anyone who can reach that host/port. The tool has no built-in authentication, so only do this on a trusted, access-controlled network (or behind a reverse proxy / VPN). Anyone who can open the page can use any API key configured on the server.

5. Getting an API key (or running key-free)

The tool brings no AI of its own — you supply access. A key is tied to a **provider**, not a single model: one Anthropic key, for example, works for every Claude model. You only need a key for the provider you actually use.

5.1 Where to get a key for each provider

Provider	Models (examples)	Where to get an API key
Anthropic (Claude)	Claude Opus 4.8, Sonnet 4.6, Haiku 4.5	https://console.anthropic.com/
OpenAI	GPT-5.x, GPT-4.1, GPT-4o, o-series	https://platform.openai.com/
Google Gemini	Gemini 3 Pro, 2.5 Pro/Flash	https://aistudio.google.com/
DeepSeek	DeepSeek Chat (V3), Reasoner (R1)	https://platform.deepseek.com/
Kimi (Moonshot)	Kimi K2, Moonshot v1	https://platform.moonshot.ai/
Local / OpenAI-compatible	Any local or proxied model	No key required — see 5.3

5.2 Do I need a key for every model?

No — one key per **provider**, and it covers all of that provider's models. If you only use Claude, load one Anthropic key and you are done. Add a second key only to switch to a different provider. The same selected model powers both the analysis and the follow-up chat.

5.3 Running key-free with a local model

1. Install and start a local runtime such as **Ollama** (or LM Studio / vLLM). For Ollama: `ollama run llama3.1`.
2. In **Step 1 (AI Model)**, choose the **Local / OpenAI-compatible** provider.
3. Set the **Base URL** (Ollama default `http://localhost:11434/v1`; LM Studio `http://localhost:1234/v1`).
4. Type the local model's id (e.g. `llama3.1`) and leave the API key blank.

> A very small local model may occasionally produce a malformed result; if that happens, re-run or choose a more capable model. Cloud models generally give the most complete analyses and diagrams.

6. Step 1 — Choose your AI model & key

The first step is choosing which AI performs the analysis and entering your API key for that provider. Pick a provider and model (or type a custom model id), then paste your key and click **Verify & continue**.

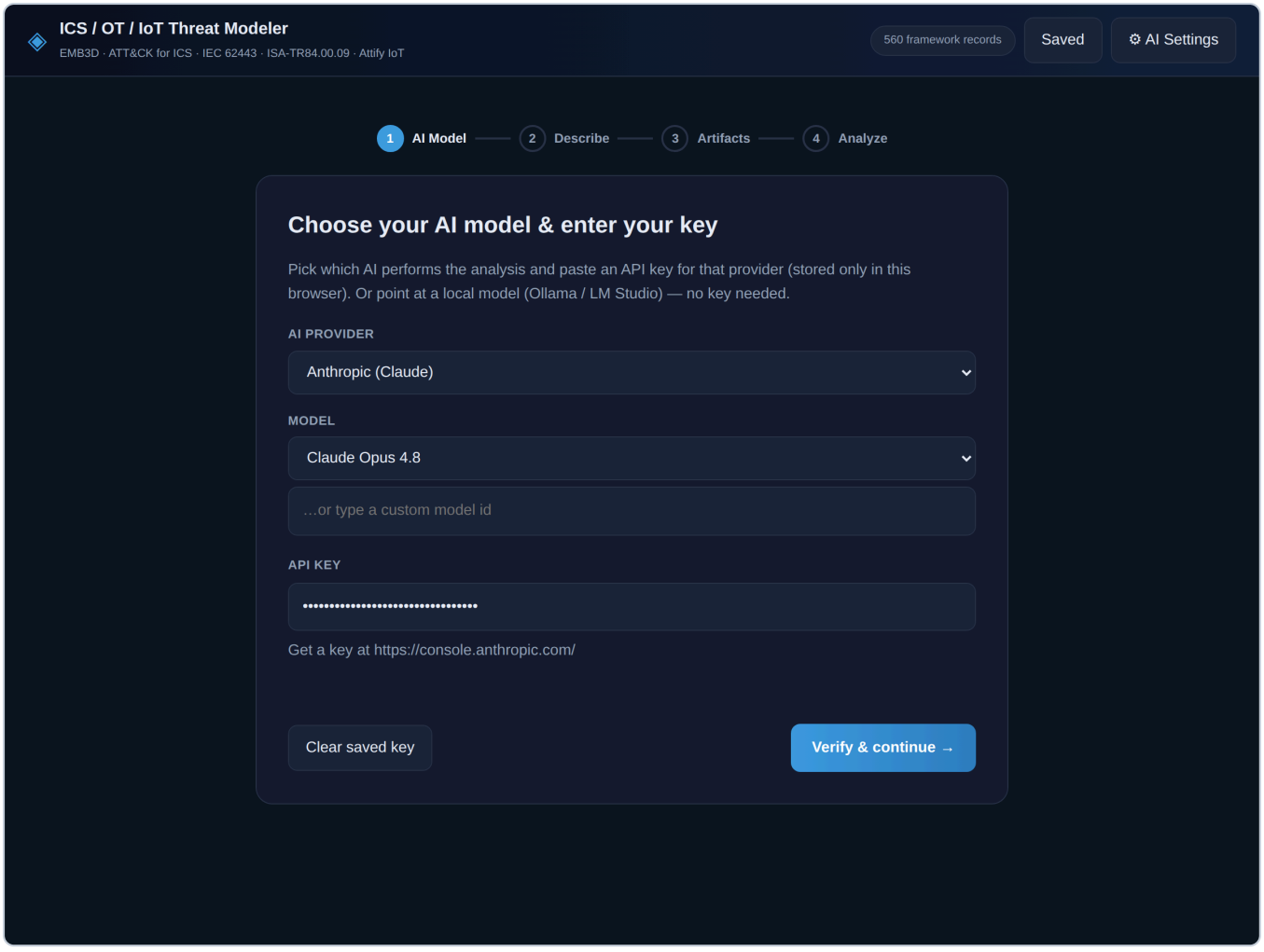


*Figure 6.1 — Step 1. Choose provider/model, paste the key, and click **Verify & continue**. Use the **Local** provider with a Base URL for a key-free local model.*

> **Your key is checked immediately — no troubleshooting. Verify & continue** makes a quick, no-cost call to the provider to confirm the key works *before* moving on. If it's good, you advance to Step 2 and the key is saved (only then). If it's rejected, you see a plain-language message right there and the bad key is **never stored**, so it can't get "stuck" and be reused on the next run.

A key that verifies is remembered in this browser, so on later runs Step 1 is pre-filled — just click **Verify & continue** again. Two controls keep you in control of the stored key:

- **Clear saved key** — wipes the stored key for the selected provider and empties the field, so you can paste a fresh one. (No cache-clearing or browser steps needed.)
- **⚙ AI Settings** (top-right) — jumps back to this step anytime to switch model or update your key.

> **Where the key comes from.** The tool checks, in order: (1) the key you verified here in Step 1, (2) the request itself, and (3) an environment variable set at launch (Section 14). You need a key only for the

one provider you use; it covers all of that provider's models.

If a key is ever rejected during analysis. On the rare chance a key stops working mid-use (e.g. it was revoked), the tool automatically clears it and returns you to Step 1 with a message — just paste a valid key and verify again. You never have to hunt through browser settings.

7. Steps 2–4 — Describe, Artifacts, Analyze

After Step 1 (AI model & key), the guided wizard walks through three more steps shown across the top: **Describe**, **Artifacts**, and **Analyze**.

7.1 Step 2 — Describe

Name the component, device, or system under evaluation. Be specific: include the model number and any notable interfaces you know. Add scope/context if useful (deployment, whether it performs a safety function, boundaries, assumptions). Click **Continue**.

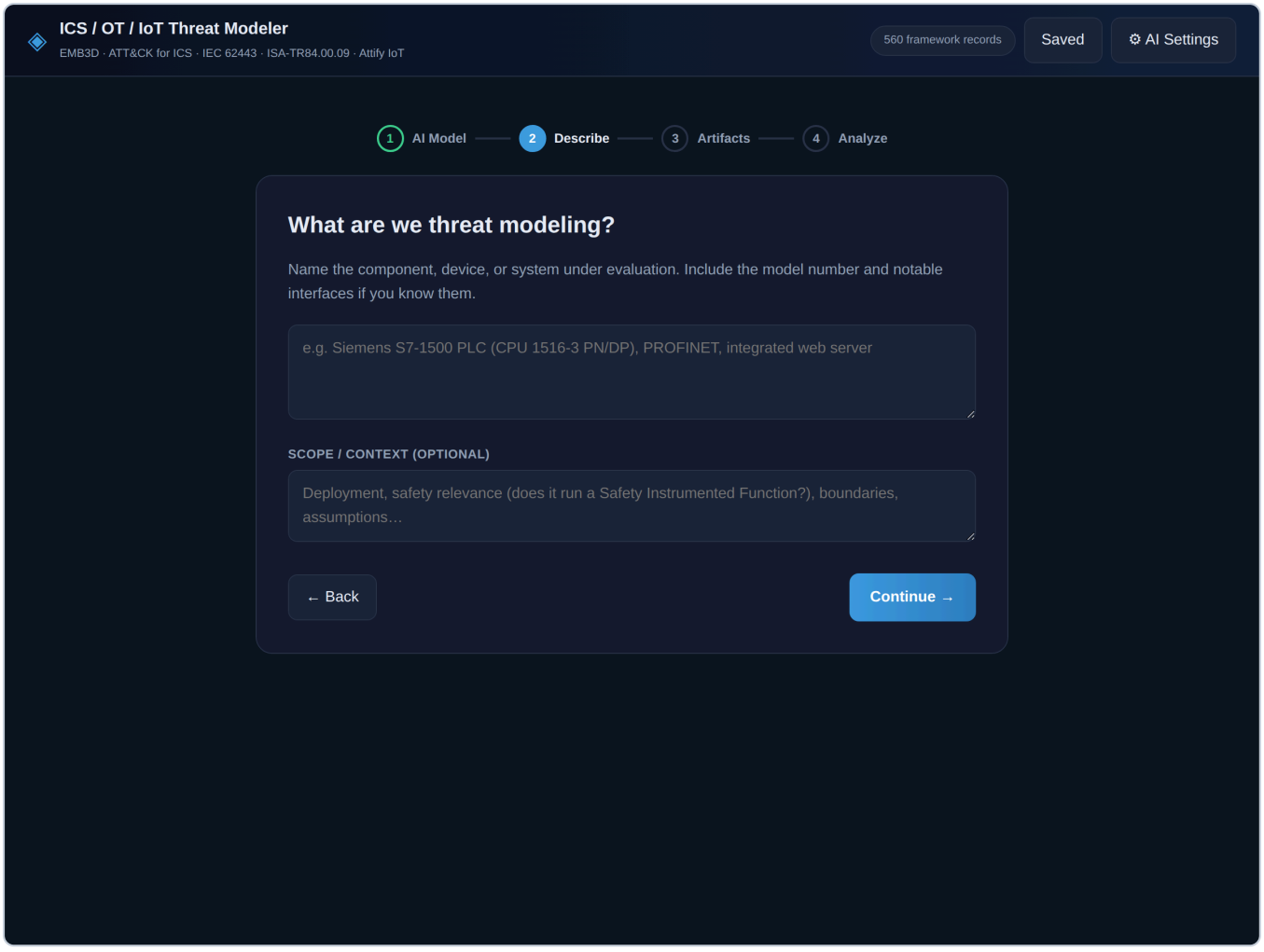


Figure 7.1 — Step 2. Name the target and optionally add scope. This is the only information the tool strictly needs.

> **More detail in, better model out.** A model number plus the interfaces and protocols the device exposes yields far more specific results than a bare product name — and attaching a datasheet (Step 2) improves it further.

7.2 Step 3 — Artifacts

Optionally upload supporting artifacts. Drag & drop files or click to browse. The AI reads these to model the *real* device.

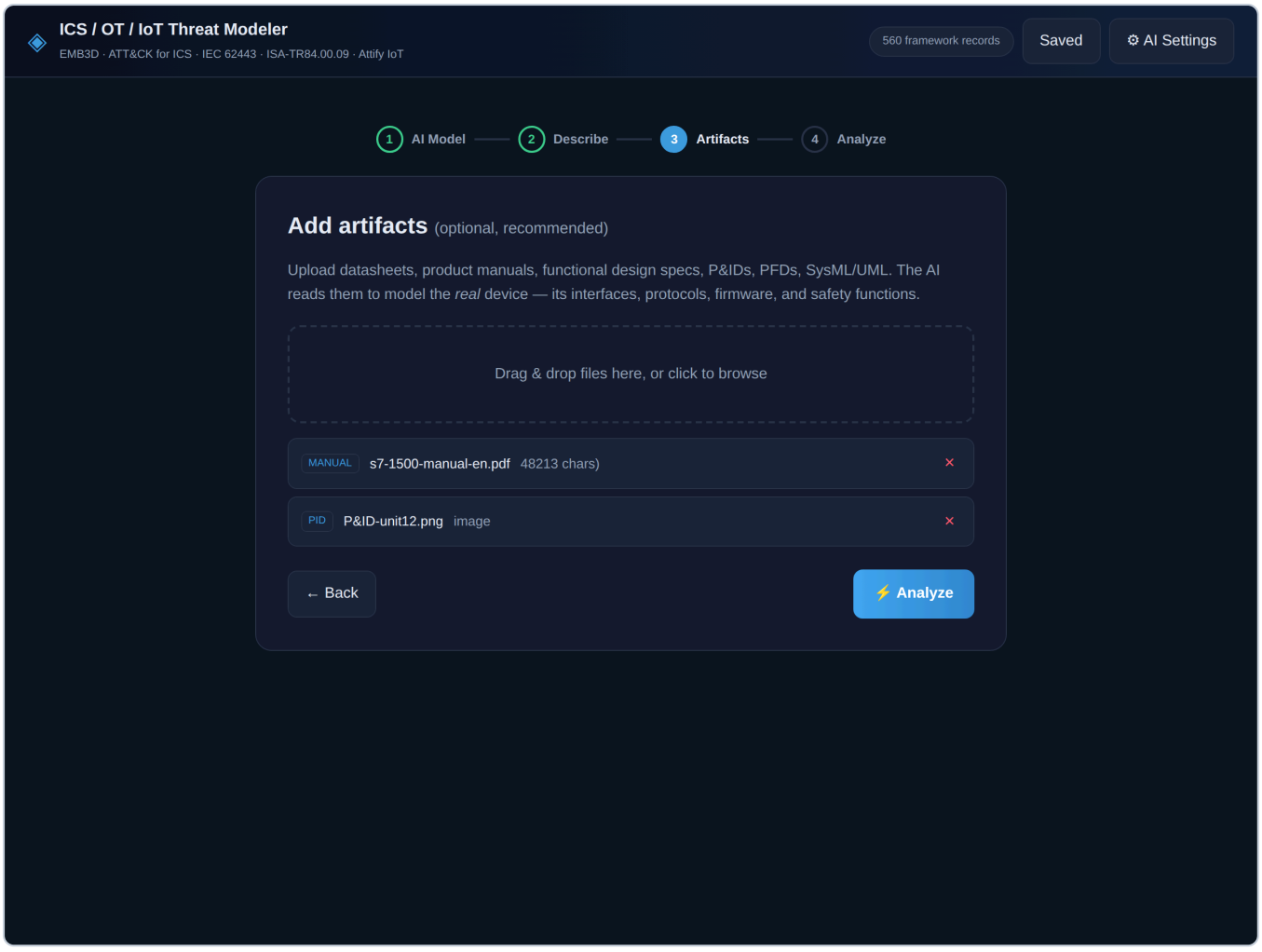


Figure 7.2 — Step 3. Uploaded artifacts are listed with their detected type and extracted size. Remove any with the × button.

Artifact type	How it is used
Datasheets, manuals, functional design specs (PDF / DOCX / text)	Text is extracted (using the standard library — no installs) and added to the prompt so the AI can cite real properties, interfaces, and protocols.
P&IDs / PFDs (PDF or image)	Text is extracted where present; drawings supplied as images are sent to a vision-capable model to read the diagram.
SysML / UML (text/XMI exports or images)	Text exports are parsed; diagram images are sent to a vision model.
Network / architecture diagrams, PCB photos (images)	Sent to a vision-capable model for interpretation.

> **Vision for drawings.** To get the most from image artifacts, pick a vision-capable model (Claude, GPT, or Gemini) in AI Settings. A scanned drawing with no embedded text is best attached as an image so a vision model can read it.

Then click ⚡ **Analyze**.

7.3 Step 4 — Analyze

Click ⚡ **Analyze**. The tool contacts the model and shows live progress while the AI works. A full analysis typically takes about one to two minutes depending on the model and target. When it finishes, the results screen appears automatically. (If the API key is missing or rejected at this point, the tool returns you to Step 1 to fix it — see Section 6.)

8. What the AI does automatically

You do not choose frameworks, properties, threats, techniques, or requirements — the AI does all of that. On each run it:

- Examines the target against **every** embedded framework (EMB3D, ATT&CK for ICS, IEC 62443 3-3/4-2/4-1/2-1, ISA-TR84.00.09, Attify) and selects the relevant items itself.
- **Mines your uploaded artifacts** — reading datasheets, manuals, design specs, P&IDs/PFDs, and SysML/UML (text and images) to extract the device's real components, interfaces, protocols, ports, memory/debug features, firmware/update mechanisms, credentials, and safety functions.
- Builds the system model, EMB3D device model, threat and attack mappings, 62443 requirement mappings, and the safety analysis from those facts — stating assumptions where a detail is unknown rather than inventing it.
- Produces the narrative summary, all mapping tables, a risk register, and the three diagrams.

9. Understanding the results

The results screen opens with a coloured risk summary (Critical / High / Medium / Low from the risk register), a prominent **Download full PDF report** button, and the full report rendered below.

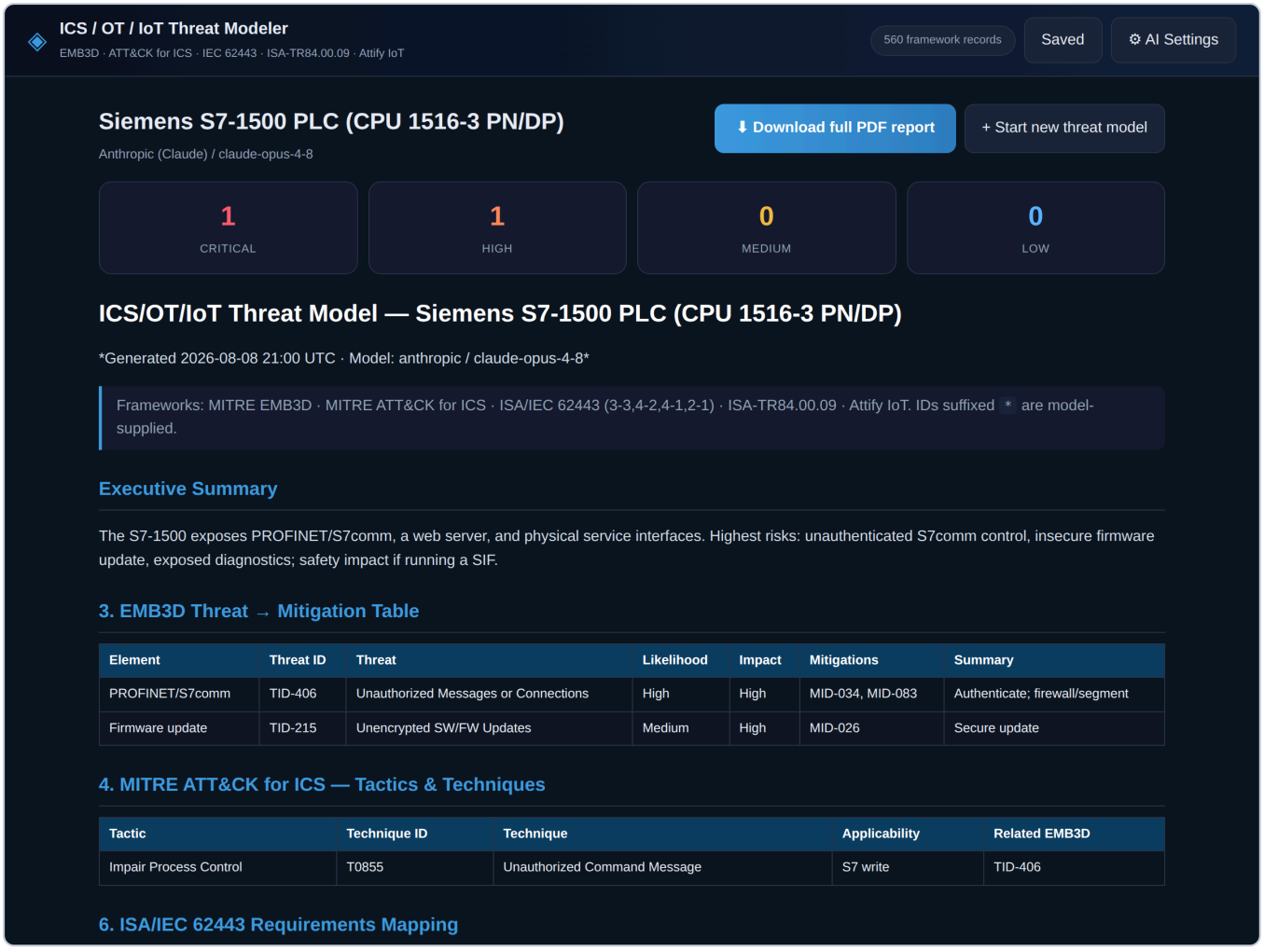


Figure 9.1 — The results screen: risk summary, download button, and the rendered report with mapping tables. (Example content.)

Section	What it contains
Executive Summary	A short, risk-focused overview for management.
System Model	Components/sub-components, interfaces, data flows across trust boundaries, and 62443 zones/conduits.
EMB3D Device Model	The device's EMB3D property types with the concrete features/interfaces present.
EMB3D Threat → Mitigation Table	Applicable EMB3D threats (TID) with likelihood, impact, and EMB3D mitigations (MID).
ATT&CK for ICS	Relevant adversary tactics and techniques (T0xxx), how each applies, and related EMB3D threats.
Attify Attack Surface	Reverse-engineering / testing vectors across hardware, firmware, radio, network, and applications.

IEC 62443 Mapping	Component (4-2), system (3-3), lifecycle (4-1), and asset-owner program (2-1) requirements addressing the findings.
ISA-TR84.00.09 Safety	Whether a safety function is involved, and cyber-initiated hazardous scenarios with attack path, consequence, severity, and countermeasures.
Risk Register	Consolidated risks with likelihood, impact, level, recommendation, and mapped requirements.

> **Reading the IDs.** Identifiers come straight from the frameworks (e.g. `TID-406`, `T0855`, `CR 3.1`, `SR 5.1`). Any identifier ending in `*` was supplied by the model rather than the embedded reference set — verify those against the primary source if precision matters.

10. The diagrams

Below the report, the tool renders three diagrams — generated entirely by the tool (no external libraries) and included in the PDF.

Diagram	Shows
Threat Model Diagram	Components grouped into security zones, external systems, data flows between them, and annotated attacker entry points (dashed, with the relevant threat ID).
ICS Cyber Kill Chain	The attack progression through the relevant ATT&CK-for-ICS tactics (Initial Access → … → Impact), with the mapped techniques at each stage.
Attack Mind Map	The target branched by attack-surface domain (Hardware, Firmware, Radio, Network, Application) with the specific vectors and their framework IDs.

ICS / OT / IoT Threat Modeler

EMB3D · ATT&CK for ICS · IEC 62443 · ISA-TR84.00.09 · Attify IoT

560 framework records | Saved | ⚙ AI Settings

Siemens S7-1500 PLC (CPU 1516-3 PN/DP)

Anthropic (Claude) / claude-opus-4-8

⬇ Download full PDF report | + Start new threat model

1	1	0	0
CRITICAL	HIGH	MEDIUM	LOW

ICS/OT/IoT Threat Model — Siemens S7-1500 PLC (CPU 1516-3 PN/DP)

Generated 2026-08-08 21:00 UTC · Model: anthropic / claude-opus-4-8

> Frameworks: MITRE EMB3D · MITRE ATT&CK for ICS · ISA/IEC 62443 (3-3,4-2,4-1,2-1) · ISA-TR84.00.09 · Attify IoT. IDs suffixed `*` are model-supplied.

Executive Summary

The S7-1500 exposes PROFINET/S7comm, a web server, and physical service interfaces. Highest risks: unauthenticated S7comm control, insecure firmware update, exposed diagnostics; safety impact if running a SIF.

3. EMB3D Threat → Mitigation Table

Element	Threat ID	Threat	Likelihood	Impact	Mitigations	Summary
PROFINET/S7comm	TID-406	Unauthorized Messages or Connections	High	High	MID-034, MID-083	Authenticate; firewall/segment
Firmware update	TID-215	Unencrypted SW/FW Updates	Medium	High	MID-026	Secure update

4. MITRE ATT&CK for ICS — Tactics & Techniques

Tactic	Technique ID	Technique	Applicability	Related EMB3D
Impair Process Control	T0855	Unauthorized Command Message	S7 write	TID-406

6. ISA/IEC 62443 Requirements Mapping

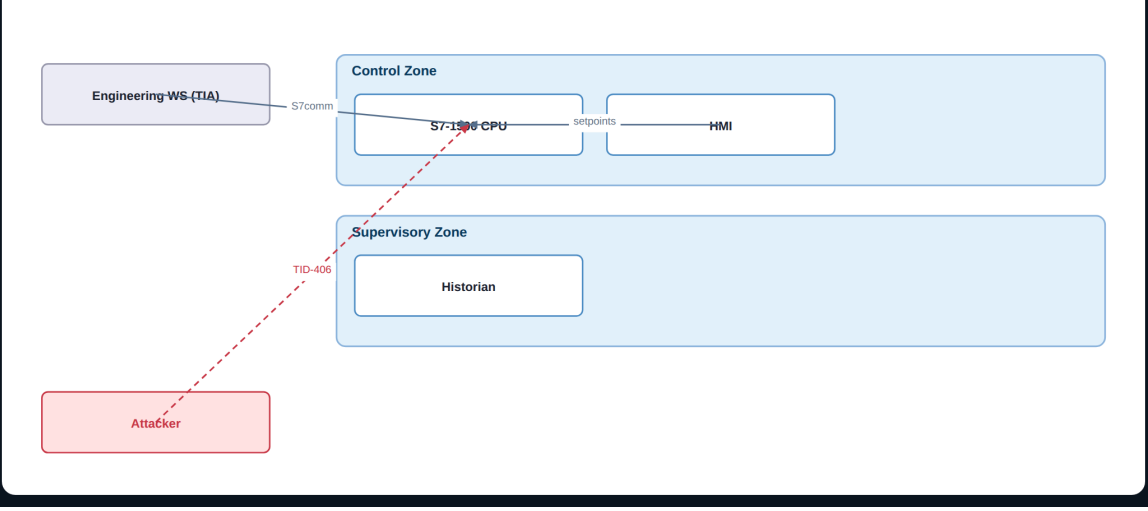
6.1 62443-4-2 — Component Requirements

Req ID	Name	Design Mitigation	Addresses Threats
CR 3.1	Communication integrity	Authenticate S7comm	TID-406

8. Risk Register

ID	Risk	Threats	ATT&CK	Likelihood	Impact	Risk	Recommendation	Requirements
R1	Unauth S7comm control	TID-406	T0855	High	High	Critical	Segment + authenticate	CR 3.1
R2	Insecure firmware	TID-215	T0857	Medium	High	High	Signed firmware	CR 3.4

Threat Model Diagram



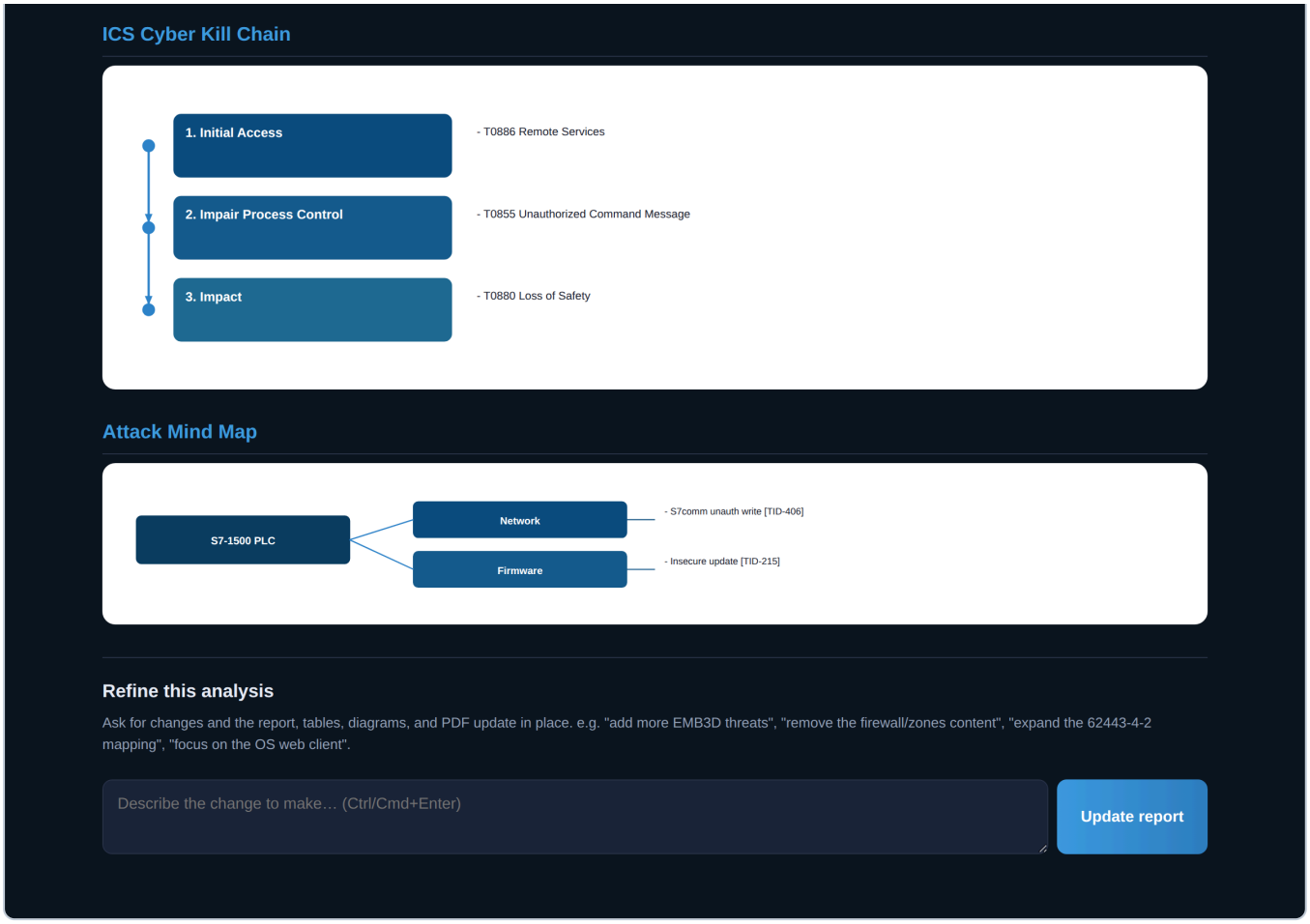


Figure 10.1 — The results screen showing the report followed by the threat-model diagram, ICS cyber kill chain, and attack mind map, plus the "Refine this analysis" box and "Start new threat model" button. (Example content.)

11. The PDF report

Click **Download full PDF report** to get a single, self-contained PDF. It is generated by the tool itself (no browser print step, no external software) and contains, in order:

- Title and generation details, with a framework note.
- The executive summary (narrative).
- The system model tables (components, interfaces, data flows, zones/conduits).
- The EMB3D device-model table.
- All mapping tables: EMB3D threats → mitigations, ATT&CK for ICS, Attify surface, IEC 62443 (4-2, 3-3, 4-1, 2-1), and the ISA-TR84.00.09 safety analysis.
- The risk register.
- All three diagrams (threat model, ICS kill chain, attack mind map) as vector graphics.
- Assumptions and references (including the artifact filenames used).

> **Sharing.** The PDF is a complete deliverable — hand it to a colleague or attach it to a design review. You can also export the underlying data: use the Saved panel to re-download the PDF for any past report.

12. Refining the analysis (updates the report & PDF)

Beneath the results is a **Refine this analysis** box. Type the change you want and the tool re-runs the model and **updates the on-screen report, all mapping tables, the three diagrams, and the downloadable PDF in place** — you don't start over or lose anything. The refine uses the same provider/model you already set up.

> **What "refine" does.** Your instruction is applied to the current model, and the complete report is regenerated and saved over the same record. Because the PDF is produced from that saved record, **Download full PDF report** immediately reflects your change. Use it to add depth, remove content you didn't want, or re-focus the analysis.

Example refinements

- "Add more EMB3D threats for the engineering station and historian."
- "Remove the firewall/zones content and focus on component vulnerabilities."
- "Expand the 62443-4-2 mapping and add the relevant 4-1 practices."
- "Add a hazard scenario for loss of the safety function."

After a refine completes, a confirmation appears and the report, diagrams, and PDF are all current. You can refine repeatedly.

13. Saved reports, new models & comparing components

13.1 Saved reports

Every analysis is saved automatically to a local database. Click **Saved** (top-right) to open the panel, where you can **Open** a past report (report, diagrams and all), download its **PDF**, or **Delete** it.

> Saved reports live in a local file (default `~/.ics_threat_modeler.db` ; change with `--db`). They contain your reports only — never your API keys.

13.2 Starting another threat model

When you finish a component and want to model another, click **+ Start new threat model** on the results screen — no browser refresh needed. If your API key is already set up, the tool asks whether to keep the same model or switch:

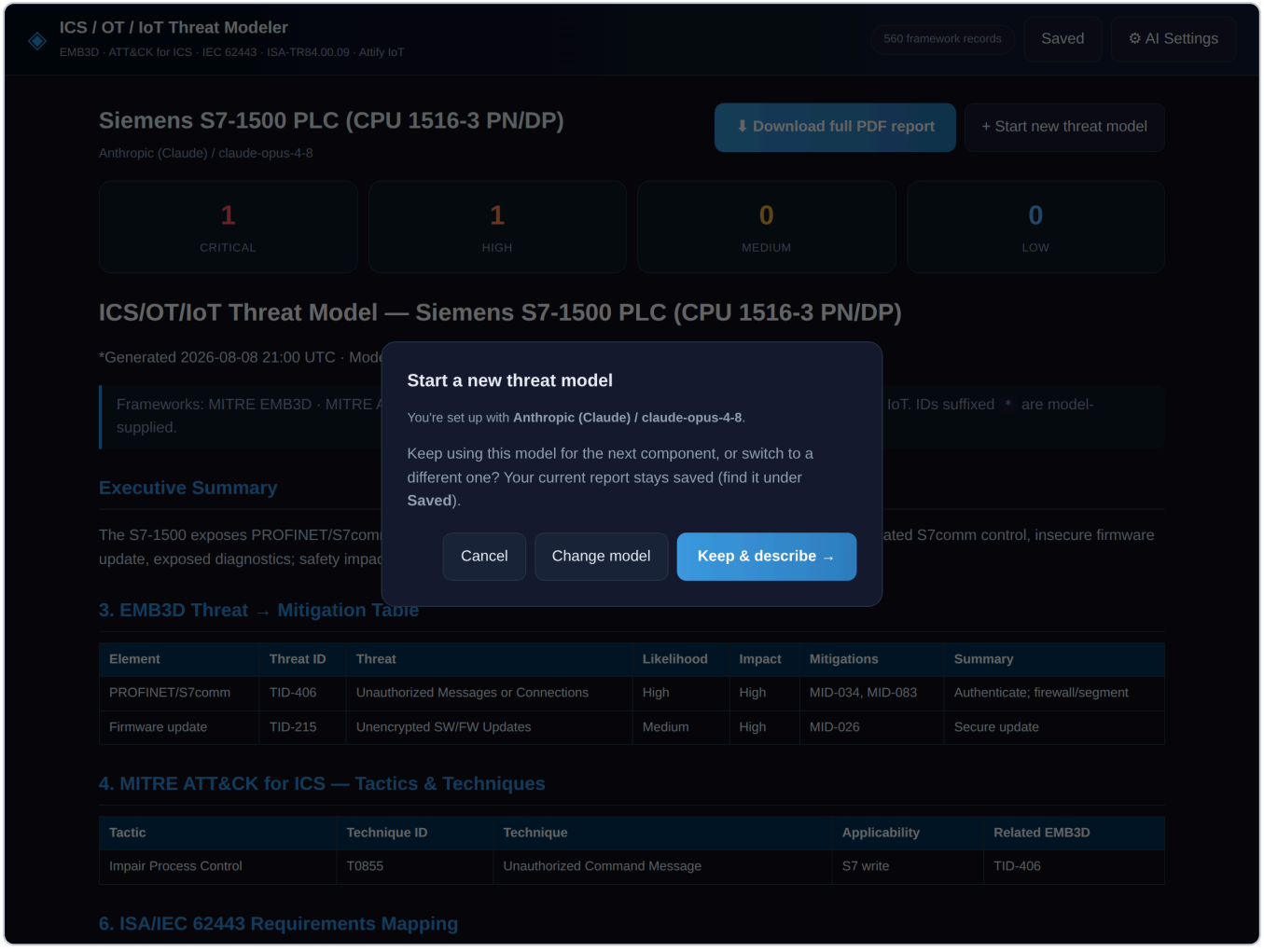


*Figure 13.1 — Starting a new threat model. **Keep & describe** jumps straight to the Describe step with your model already configured; **Change model** returns to Step 1 to pick a different provider/model; your current report stays saved.*

This keeps the flow smooth across many components: analyze → refine → start new → describe the next component, without re-entering your key each time.

13.3 Comparing components

To compare several components side by side, open **Saved**, tick two or more reports, and click **Compare selected**. The tool shows a comparison of risk posture across them.

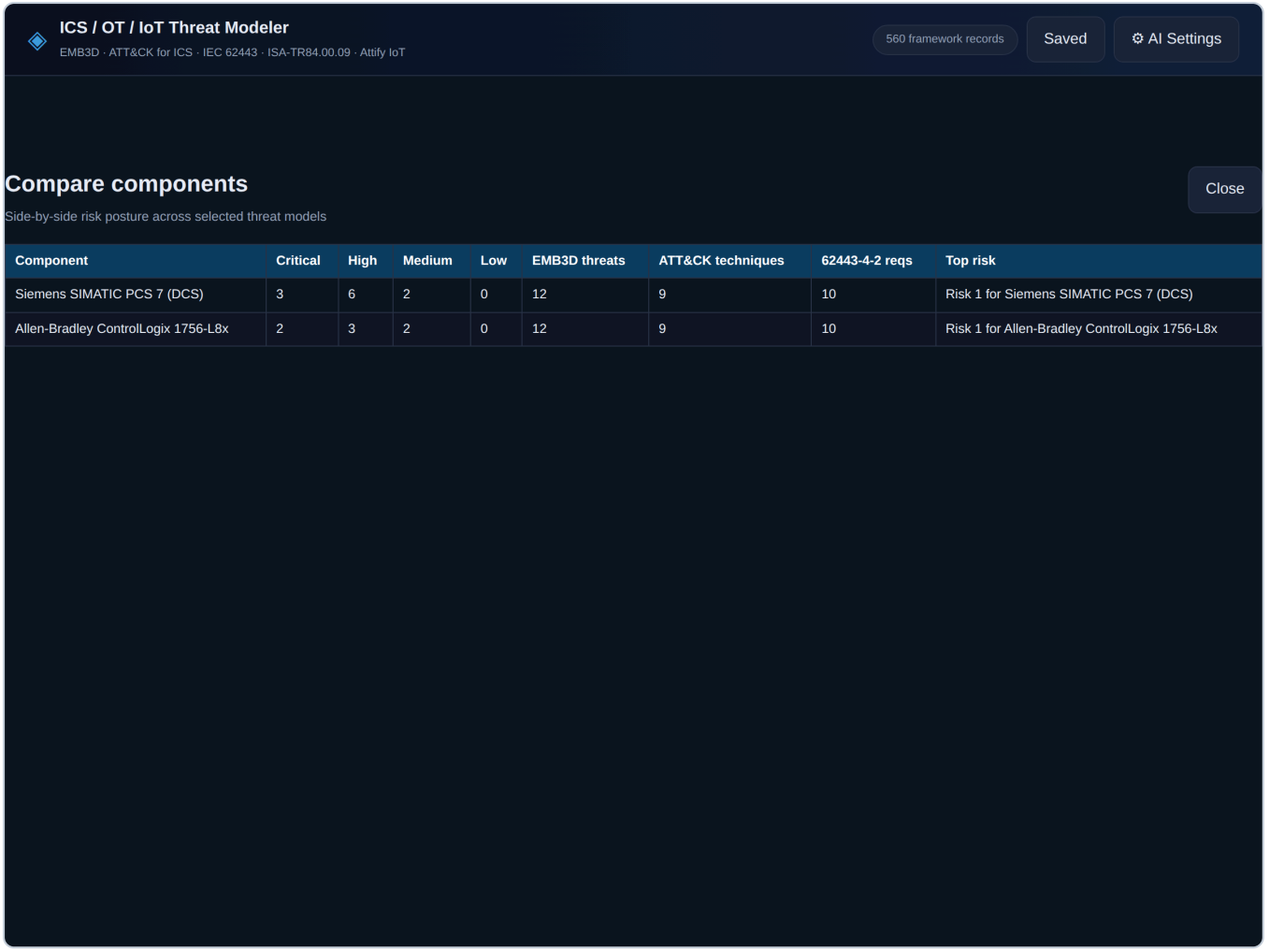


Figure 13.2 — Compare view: each component's Critical/High/Medium/Low counts, number of EMB3D threats, ATT&CK-for-ICS techniques, and 62443-4-2 requirements, plus its top risk — useful for prioritising across a fleet or a procurement shortlist.

14. Command-line & environment reference

Command-line options

Option	Default	Description
`--host HOST`	127.0.0.1	Address to bind. Use `0.0.0.0` to allow other machines (see Section 4 warning).
`--port PORT`	8000	Port to serve on.
`--db PATH`	~/.ics_threat_modeler.db	Location of the saved-reports database.
`--open`	off	Open the UI in your default browser on start.
`--stats`	—	Print embedded framework record counts and exit.
`-h`, `--help`	—	Show usage help.

Environment variables

Variable	Purpose
`ANTHROPIC_API_KEY`	Server-side key for Anthropic (Claude).
`OPENAI_API_KEY`	Server-side key for OpenAI.
`GOOGLE_API_KEY`	Server-side key for Google Gemini.
`DEEPSEEK_API_KEY`	Server-side key for DeepSeek.
`MOONSHOT_API_KEY`	Server-side key for Kimi (Moonshot).
`LOCAL_OPENAI_BASE_URL` / `LOCAL_OPENAI_MODEL`	Default base URL and model for the local provider.
`ITM_HOST` / `ITM_PORT` / `ITM_DB`	Alternate way to set host, port, and database path.

15. Troubleshooting

Symptom	Cause & fix
"Your API key was rejected" at Step 1	The key check failed. Re-paste the full key (Anthropic keys start with `sk-ant-`) with no extra spaces, confirm it's active in your provider console, then click **Verify & continue**. Nothing is stored until it verifies.
Need to change or remove a saved key	Go to Step 1 (or click ⚙ **AI Settings**) and use **Clear saved key**, then paste a new one and verify. No browser cache-clearing required.
An error mentioning 404 / model not found	The model id is not valid for that provider. Pick one from the dropdown or type the exact current id in the custom-model box.
"Could not parse model JSON"	The model produced malformed output (more common with very small local models). Re-run or choose a more capable model.
A PDF/DOCX artifact shows a warning that no text was found	The file is likely a scanned or vector-only document. Export the drawing to an image and attach that so a vision model can read it.
"Address already in use" on start	Another process holds the port. Start with a different `--port`, e.g. `--port 8090`.
Local model connection fails	Confirm the local runtime is running and the Base URL is correct (Ollama: `http://localhost:11434/v1`) and the model is pulled/loaded.
Analysis is slow	Large "thinking"/pro models take longer. A full analysis is typically 1–2 minutes; try a faster model for quick passes.

16. Frameworks, attribution & disclaimer

This tool references and maps to third-party frameworks that remain the property of their respective owners. The embedded reference data consists of identifiers, titles, and concise paraphrased descriptions provided for interoperability and educational use; it is not the official, complete text of any standard. For authoritative content, consult the primary sources:

- MITRE EMB3D™ — https://emb3d.mitre.org
- MITRE ATT&CK® for ICS — https://attack.mitre.org/matrices/ics/
- ISA/IEC 62443 series — https://www.isa.org (standards available for purchase)
- ISA-TR84.00.09 — https://www.isa.org (technical report available for purchase)
- Attify IoT attack-surface methodology — https://www.attify.com

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> **Professional judgement required.** This tool assists expert threat modeling; it does not replace a qualified assessment. AI-generated content can contain errors or omissions. Validate findings — especially model-supplied identifiers marked with `*` — against primary sources and your own engineering analysis before relying on them for design, procurement, or safety decisions.

17. Frequently asked questions

Do I have to install anything?

No. Only Python 3.9+ is required. The tool uses the standard library for everything, including reading PDF/Word artifacts and producing the PDF report.

Do I need an internet connection?

You need outbound access to whichever AI provider you choose. With a local model the tool runs fully offline. The report and diagrams render locally either way.

Do I need a separate key for each model?

No — one key per **provider** covers all of that provider's models. Add more keys only to switch providers.

Do I have to pick the frameworks or threats myself?

No. You just name the target and (optionally) upload artifacts. The AI selects all relevant properties, threats, techniques, and requirements and builds the diagrams.

Is my data sent anywhere I didn't choose?

Your target, artifacts, and reports stay on your machine. The only outbound content is the prompt sent to the AI provider you selected. Keys are stored in your browser or environment and never written to the saved-reports database.

Where are my saved reports stored?

In a local database file (default `~/.ics_threat_modeler.db`; change with `--db`). Delete individual reports from the Saved panel.

How current is the framework data?

The embedded catalogues reflect the frameworks at the time this edition was built (560 records; run `--stats` to see counts). For the latest revisions consult the primary sources in Section 16; in hybrid mode the model can also supplement with newer items (marked `*`).

18. Quick-reference card

Task	How
Start the tool	`python3 ics_threat_modeler.py --open`
Use a different port	`--port 8090`
Set a key at launch	`ANTHROPIC_API_KEY=sk-... python3 ics_threat_modeler.py`
Choose model & key in the UI	Step 1 (AI Model) → provider + key → Continue
Use a local model	Step 1 → Local; Base URL `http://localhost:11434/v1` ; no key
Run an analysis	Step 1 model+key → Step 2 name it → Step 3 artifacts → Step 4 Analyze
Get the PDF	Results screen → Download full PDF report
Refine (updates report + PDF)	Results screen → Refine this analysis box
Start another component	Results screen → + Start new threat model → Keep model or Change
Compare components	Saved → tick 2+ → Compare selected
Reopen past work	Saved → Open (or PDF)
See record counts	`python3 ics_threat_modeler.py --stats`
Stop the tool	Ctrl+C in the terminal

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This tool assists but does not replace qualified security and safety analysis.